Module 31

Application Design and Development/1: Architecture

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Week Recap

- Studied the Normal Forms and their Importance in Relational Design - how progressive increase of constraints can minimize redundancy in a schema

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- Studied the Normal Forms and their Importance in Relational Design - how progressive increase of constraints can minimize redundancy in a schema
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- Understood multi-valued dependencies to handle attributes that can have multiple values
- Learnt Fourth Normal Form and decomposition to 4NF
- Discussed aspects of the database design process
- Studied the issues with temporal data

Module Objectives

Goals

- What are the Application Programs across various sectors?

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- Commonality of architecture across applications

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- What are the Application Programs across various sectors?
- Commonality of architecture across applications
- Understanding the classification and evolution of the architectures

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Goals

- What are the Application Programs across various sectors?
- Commonality of architecture across applications
- Understanding the classification and evolution of the architectures
- A look at the architecture for a few sample applications

Module Outline

- Application Programs

Module Outline

- Application Programs
- Application Architecture with classification and evolution

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- Application Programs
- Application Architecture with classification and evolution
- Sample application architectures

Application Programs and Architecture

Application Programs: Internet / Web or Mobile

- **Financial:**

Application Programs: Internet / Web or Mobile

- **Financial:**
  - Netbanking: SBI, PNB, BoB, Canara, HDFC, ICICI

Application Programs: Internet / Web or Mobile

- **Financial:**
  - Netbanking: SBI, PNB, BoB, Canara, HDFC, ICICI
  - Insurance & Investment: LICI, PolicyBazaar, NSDL, NPS,

Application Programs: Internet / Web or Mobile

- **Financial:**
  - Netbanking: SBI, PNB, BoB, Canara, HDFC, ICICI
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Application Programs: Internet / Web or Mobile

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  - Live Interaction: Zoom, Google Meet, Teams, Webex, Skype,

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- **Communication:**
  - Live Interaction: Zoom, Google Meet, Teams, Webex, Skype,
  - Mail: Gmail, Yahoo, Hotmail, Rediffmail, Enterprise Mail,
  - Intermittent Interaction: WhatsApp, Telegram, Signal, Skype

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  - Social Media: Facebook, Instagram, Twitter, YouTube,

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- **Knowledge Discovery:**

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- **Library:**
  - Digital Library: National Digital Library of India,
  - Archives: Internet Archive, arXiv, Nextpoint,
- **Education:**

Application Programs: Internet / Web or Mobile (cont.)

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  - Archives: Internet Archive, arXiv, Nextpoint,
- **Education:**
  - eLearning: BYJU's, IGNOU, NIIT, Edukart,
  - MOOCs: SWAYAM, edX, Coursera, Udemy,

Application Programs: Internet / Web or Mobile (cont.)

- **Document Processing:**

Application Programs: Internet / Web or Mobile (cont.)

- **Document Processing:**
  - Editing: Overleaf, Google Docs, Spreadsheet

Application Programs: Internet / Web or Mobile (cont.)

- **Document Processing:**
  - Editing: Overleaf, Google Docs, Spreadsheet
  - Website, Blog: Google Sites, WordPress, Webly,

Application Programs: Internet / Web or Mobile (cont.)

- **Document Processing:**
  - Editing: Overleaf, Google Docs, Spreadsheet
  - Website, Blog: Google Sites, WordPress, Webly,
- **Health:**

Application Programs: Internet / Web or Mobile (cont.)

- **Document Processing:**
  - Editing: Overleaf, Google Docs, Spreadsheet
  - Website, Blog: Google Sites, WordPress, Webly,
- **Health:**
  - Telemedicine: MDLIVE, Doctor on Demand,

Application Programs: Internet / Web or Mobile (cont.)

- **Document Processing:**
  - Editing: Overleaf, Google Docs, Spreadsheet
  - Website, Blog: Google Sites, WordPress, Webly,
- **Health:**
  - Telemedicine: MDLIVE, Doctor on Demand,
  - National: Aarogy Setu, CoWin, NACO App,

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- **Health:**
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- **Organizational ERP: (Intranet)**

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  - Manufacturing: Suppliers, Inventory, Customers,

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  - Hospital: Patient, Doctor, OPD, IPD, Pharmacy,

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  - Hospital: Patient, Doctor, OPD, IPD, Pharmacy,
  - Bank: Customers, Accounts, Locker, Deposits,

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  - Hospital: Patient, Doctor, OPD, IPD, Pharmacy,
  - Bank: Customers, Accounts, Locker, Deposits,
  - Courier: Customers, Parcels, Delivery Agents,

Characteristic of Application Programs

- **Diversity**: These applications widely differ in their

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  - Domain, functionality, user base, response time, scale, daily hit and many more

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- **Unity**: Yet, these have a lot in common

Characteristic of Application Programs

- **Diversity**: These applications widely differ in their
  - Domain, functionality, user base, response time, scale, daily hit and many more
- **Unity**: Yet, these have a lot in common
  - Most use an RDBMS like Oracle, DB2 MySQL, PostgreSQL, etc. for managing data

Characteristic of Application Programs

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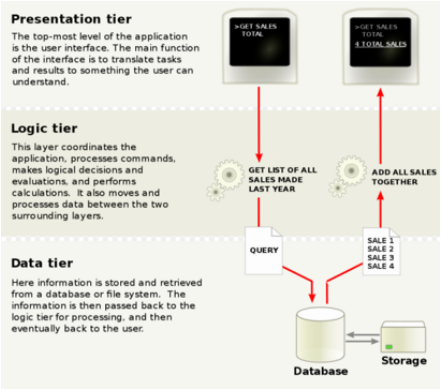
Characteristic of Application Programs (2): Architecture



Source: https://en.wikipedia.org/wiki/Multitier_architecture

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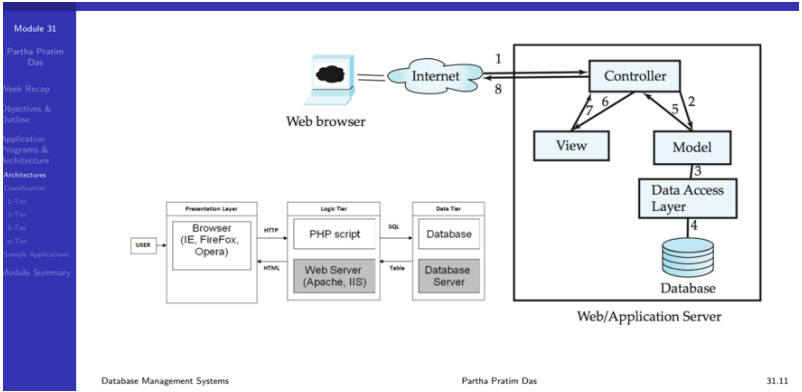

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Application Architecture (6): Data Access Layer

- Issues of modeling and design of databases have already discussed in depth through the previous module
- Issues of accessing and updating data from application will be discussed later this with through the interactions of native languages and SQL

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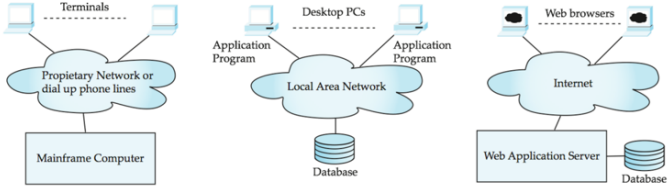
Architecture Evolution

- Three distinct eras of application architecture
  - Mainframe (1960's and 70's)
  - Personal computer era (1980's)
  - Web / Mobile era (1990's onwards)



(a) Mainframe Era (b) Personal Computer Era (c) Web era

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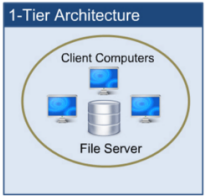

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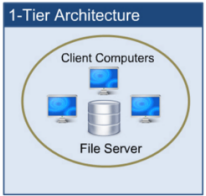

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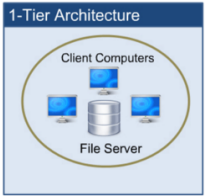
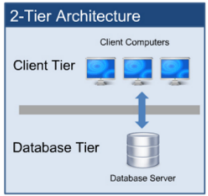


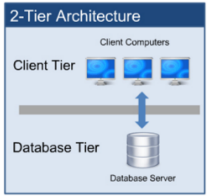
2-tier Architecture

- The two-tier is based on Client Server architecture



2-tier Architecture

- The two-tier is based on Client Server architecture
- It is like client server application



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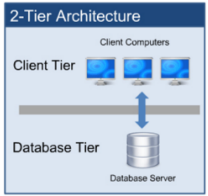
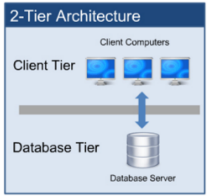


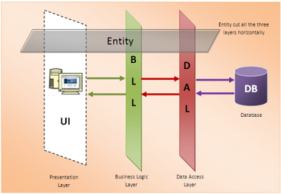
2-tier Architecture

- The two-tier is based on Client Server architecture
- It is like client server application
- The direct communication takes place between client and server
- There is no intermediate between client and server



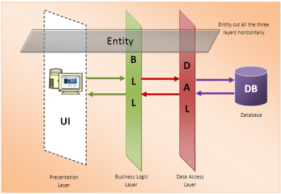
3-tier Architecture

- A 3-tier architecture separates its tiers Presentation, Logic and Data Access - from each other based on the complexity of the users and how they use the data present in the database



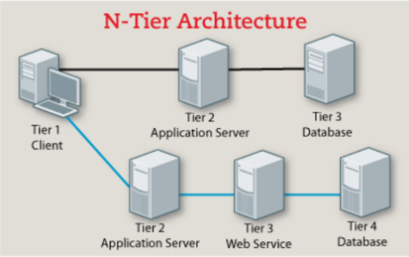
3-tier Architecture

- A 3-tier architecture separates its tiers Presentation, Logic and Data Access - from each other based on the complexity of the users and how they use the data present in the database
- It is the most widely used architecture to design a DBMS



n-tier Architecture

- An n-tier architecture distributes different components of the 3 tiers between different servers and adds interfaces tiers for interactions and workload balancing



Sample Applications in Multiple Tiers

Application	Presentation	Logic	Data	Functionality
Web Mail	Login Mail List View Inbox Sent Items Outbox Trash Mail Composer Filters	User Authentication Connection to Mail Server (SMTP, POP, IMAP) Encryption Decryption	Mail Users Address Book Mail Items	Send Receive Mails Manage Address Book
Net Banking	Login Account View Add Delete Account Add Delete Beneficiary Fund Transfer	User Authentication Beneficiary Authentication Transaction Validation Connection to Banks Gateways Encryption Decryption	Account Holders Beneficiaries Accounts Debit Credit Transactions	Check Balance and Transactions Transfer Funds
Timetable	Login Add Delete Courses, Teachers, Rooms, Slots Assignments: Teachers Course Allocations Course Room, Slots	User Authentication Timetable Assignment Logic Encryption Decryption	Courses Teachers Rooms Slots Assignments Allocations	Manage timetable for multiple courses taken by multiple teachers

Module Summary

Summary

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Module Summary

Summary

- Had a glimpse of Application Programs across various sectors
- Understood the typical architecture for an application
- Studies the classification and evolution of the architectures
- Glimpsed at architecture for a few sample applications

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